L4.2a Volumes by Slicing

Recall: solving for x, and areas of shapes

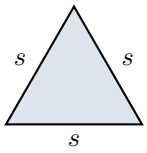
Name _____ **Solo – 9 minutes**

1. Write each curve as x in terms of y, for $x \geqslant 0$: (a) $y = x^3$ (b) $y = \sqrt{x}$

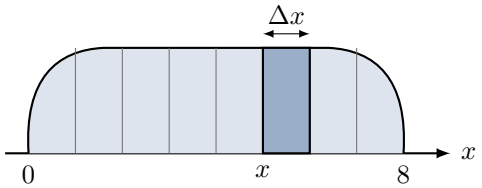
2. Write each area as a formula.

(a) A disk of radius $2 - x$.

(b) An equilateral triangle of side s.



3. A loaf of bread lies along the x-axis from $x = 0$ to $x = 8$. It is cut into n slices, each of thickness Δx. The face of the slice at x has area $A(x)$.



(a) Write the volume of one slice, approximately.

(b) Write a sum that estimates the volume of the loaf.

(c) What would make the estimate exact?